

2. Describe briefly a practical problem where calculation of stresses in a plane different from the loading direction of a structural member is useful.

This is useful in cases where there exists a joint (e.g. Glued joint, welded joint, etc.) that is not aligned with the loading direction on a structural member. This way, by using stress transformations, we would be able to find the stresses present in the plane containing the joint and be able to better predict failure.

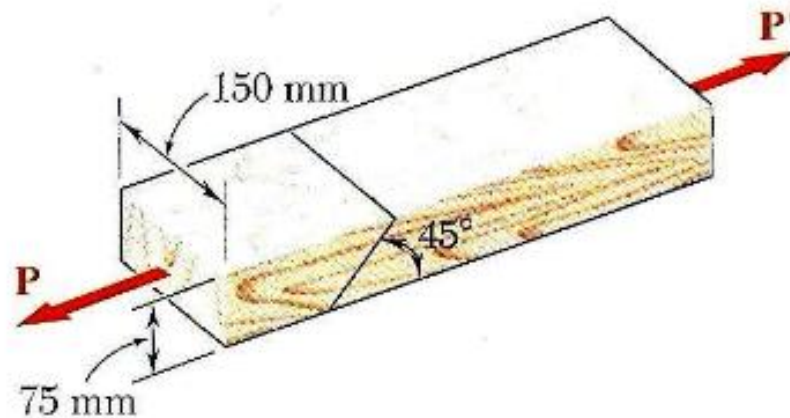


Fig. P1.29 and P1.30

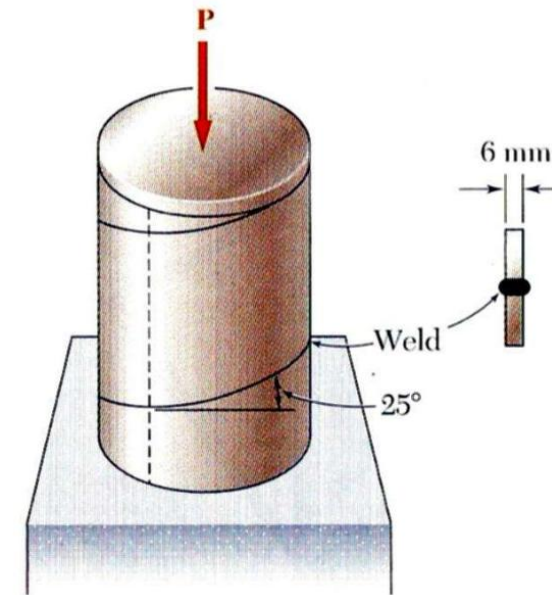


Fig. P1.33 and P1.34

3. Describe briefly a practical problem where the calculation of principle stresses is nontrivial and useful.

The principal stresses exist in an orientation when normal stresses are max, and shear stresses are zero. Trivial cases exist when the principal directions align with an applied uniaxial load. Non-trivial cases exist when a material is under combined shear and tensile stresses and the max normal stresses exist on a plane that isn't aligned with the direction of loading. This could be useful when the material or joint is weak under normal stress. For example, some glues are strong in shear but weak in tension, so we can use the principal stresses to better estimate the strength of a joint.

(Based on text Ugural and Fenster 5th edition, p1.21) A welded plate carries uniform biaxial tension $\sigma_x = \sigma$, $\sigma_y = 60 \text{ MPa}$, and $\tau_{xy} = 0$. The weld is at an angle of -20° relative to the x axis. Assuming that the stress in the weld is critical, determine the maximum applied stress σ_x if (a) the weld allowable shear stress is 30 MPa, and (b) if the weld allowable normal stress is 80 MPa. Sketch Mohr's circle for each case.

Given:

- $\sigma_y = 60 \text{ MPa}$
- $\tau_{xy} = 0 \text{ MPa}$

Find:

- Max applied stress σ_x if $\tau_{all, x'y'} = 30 \text{ MPa}$
- Max applied stress σ_x if allowable normal stress is $\sigma_{all, y'} = 80 \text{ MPa}$

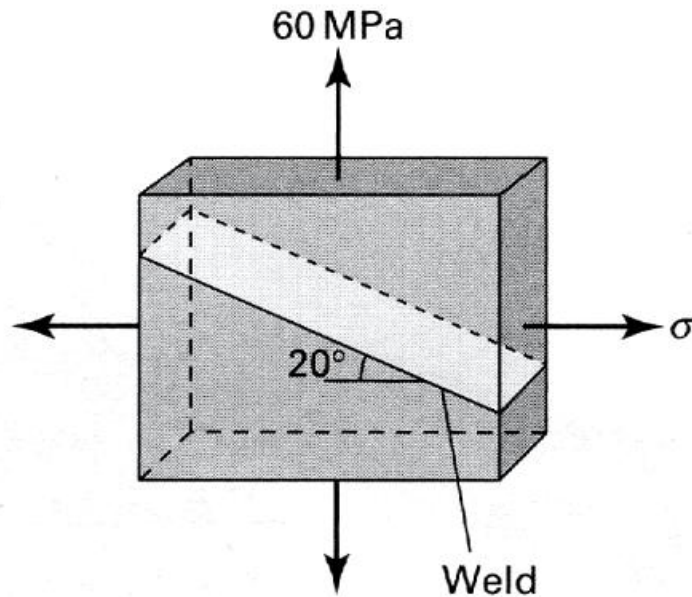
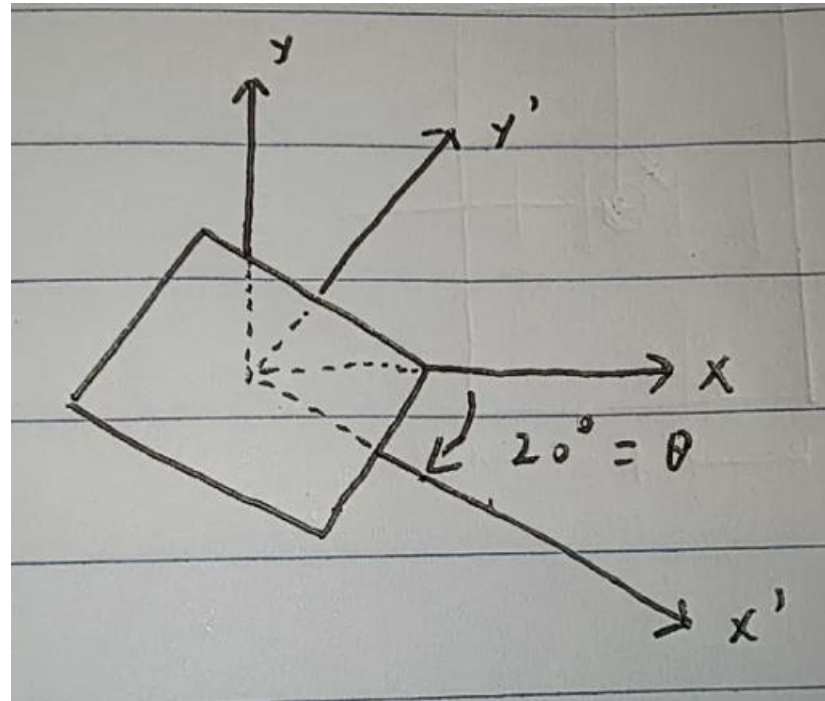


FIGURE P1.21.



Approach:

- Given $\tau_{all, x'y'} = 30 \text{ MPa}$:
 - Use the stress transformation equations to find $\Delta\sigma$ and σ_{ave} ($\tau_{xy} = 0 \text{ MPa}$):

$$\tau_{x'y'} = \tau_{xy} \cos 2\theta - \frac{\Delta\sigma}{2} \sin 2\theta$$

Where:

$$\Delta\sigma = \sigma_x - \sigma_y, \quad \sigma_{ave} = \frac{\sigma_x + \sigma_y}{2}$$

- Then use $\Delta\sigma$ and σ_{ave} to find the radius and center of Mohr's circle:

$$R = \sqrt{\tau_{xy}^2 + \left(\frac{\Delta\sigma}{2}\right)^2}, \quad C = (\sigma_{ave}, 0)$$

- Since $\tau_{xy} = 0 \text{ MPa}$, σ_x and σ_y are the principal stresses and can be found by:

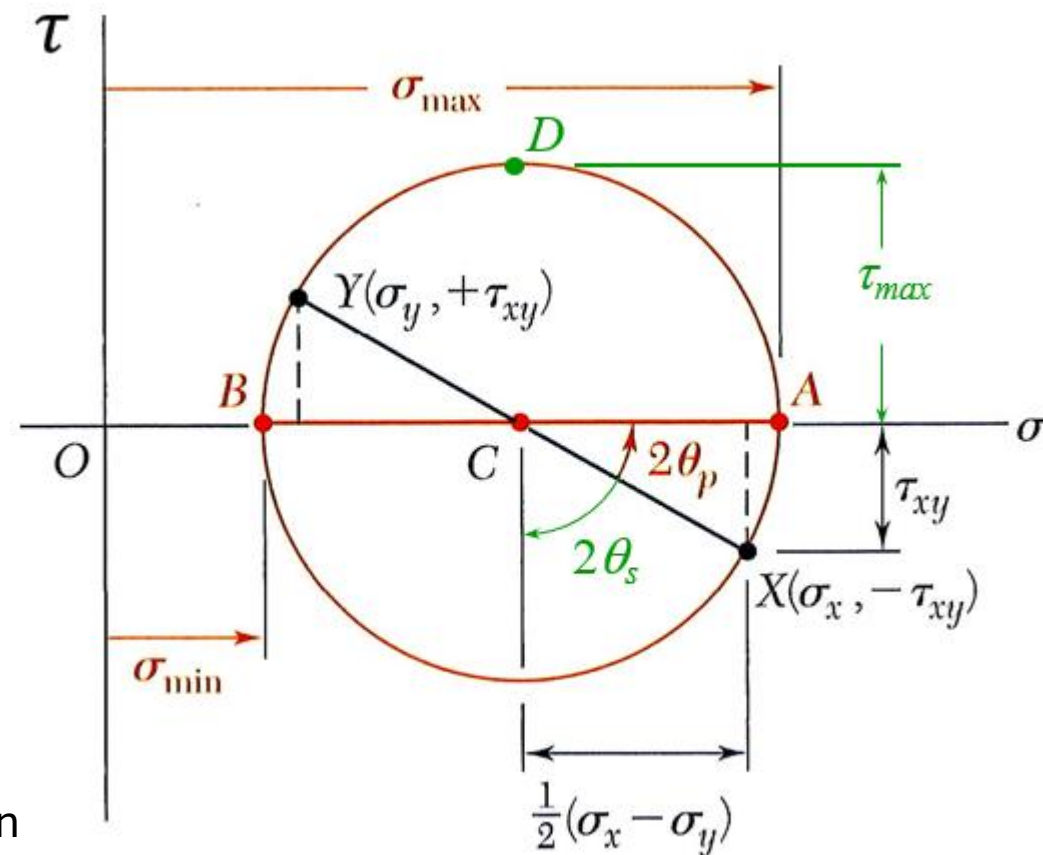
$$\sigma_{ave} + R = \sigma_x$$

Or by

$$\Delta\sigma + \sigma_y = \sigma_x$$

- Given $\sigma_{all, y'} = 80 \text{ MPa}$:
 - A similar approach to before can be taken, but we can find $\Delta\sigma$ and σ_{ave} using:

$$\sigma_{y'} = \sigma_{ave} - \frac{\Delta\sigma}{2} \cos 2\theta - \tau_{xy} \sin 2\theta$$



PART A: Given $\tau_{all, x'y'} = 30 \text{ MPa}$. Assuming $\sigma_x > \sigma_y$ to find max applied stress

$$|\tau_{x'y'}| = \left| -\frac{\Delta\sigma}{2} \sin 2\theta \right| = [30 \text{ MPa}] \longrightarrow \left| -\frac{\Delta\sigma}{2} \sin(2 \cdot -20^\circ) \right| = [30 \text{ MPa}] \longrightarrow |\Delta\sigma| = [93.3434 \text{ MPa}]$$

$$\Delta\sigma = \sigma_x - \sigma_y \longrightarrow \Delta\sigma + \sigma_y = \sigma_x$$

$$\boxed{\sigma_x = [93.3434 \text{ MPa}] + [60 \text{ MPa}] = 153.3434 \text{ MPa}}$$

Mohr's circle sketch

$$X: (\sigma_x, -\tau_{xy}) = (153.34, 0), \quad Y: (\sigma_y, \tau_{xy}) = (60, 0), \quad C: (\sigma_{ave}, 0) = (106.67, 0)$$

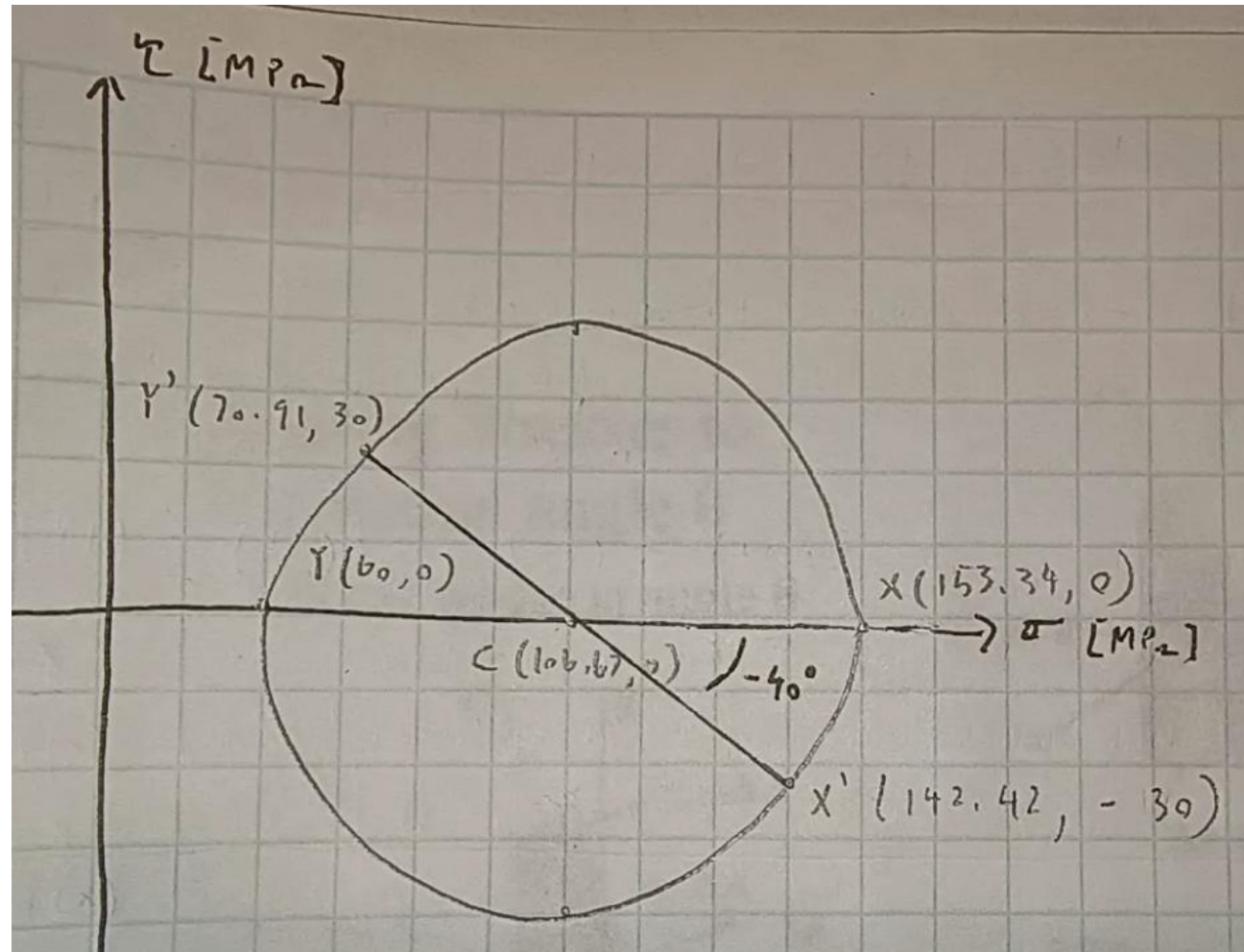
$$\sigma_{x'} = \sigma_{ave} + \frac{\Delta\sigma}{2} \cos 2\theta + \tau_{xy} \sin 2\theta = [106.67 \text{ MPa}] + \frac{[93.3434 \text{ MPa}]}{2} \cos(2 \cdot -20^\circ) = 142.42 \text{ MPa}$$

$$\sigma_{y'} = \sigma_{ave} - \frac{\Delta\sigma}{2} \cos 2\theta - \tau_{xy} \sin 2\theta = [106.67 \text{ MPa}] - \frac{[93.3434 \text{ MPa}]}{2} \cos(2 \cdot -20^\circ) = 70.91 \text{ MPa}$$

$$\tau_{x'y'} = 30 \text{ MPa}$$

PART A: Given $\tau_{all, x'y'} = 30 \text{ MPa}$. Assuming $\sigma_x > \sigma_y$ to find max applied stress

Mohr's circle sketch



PART B: Given $\sigma_{all,y'} = 80 \text{ MPa}$

$$|\sigma_{y'}| = \left| \sigma_{ave} - \frac{\Delta\sigma}{2} \cos 2\theta \right| = \left| \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta \right| = [80 \text{ MPa}]$$

$$= \left| \frac{\sigma_x}{2} (1 - \cos 2\theta) + \frac{\sigma_y}{2} (1 + \cos 2\theta) \right| = \left| \frac{\sigma_x}{2} (1 - \cos(2 \cdot -20^\circ)) + \frac{[60 \text{ MPa}]}{2} (1 + \cos(2 \cdot -20^\circ)) \right| = [80 \text{ MPa}]$$

$$\boxed{\sigma_x = 230.972 \text{ MPa}}$$

Mohr's circle sketch

$$X: (\sigma_x, -\tau_{xy}) = (230.97, 0), \quad Y: (\sigma_y, \tau_{xy}) = (60, 0), \quad C: (\sigma_{ave}, 0) = (145.48, 0)$$

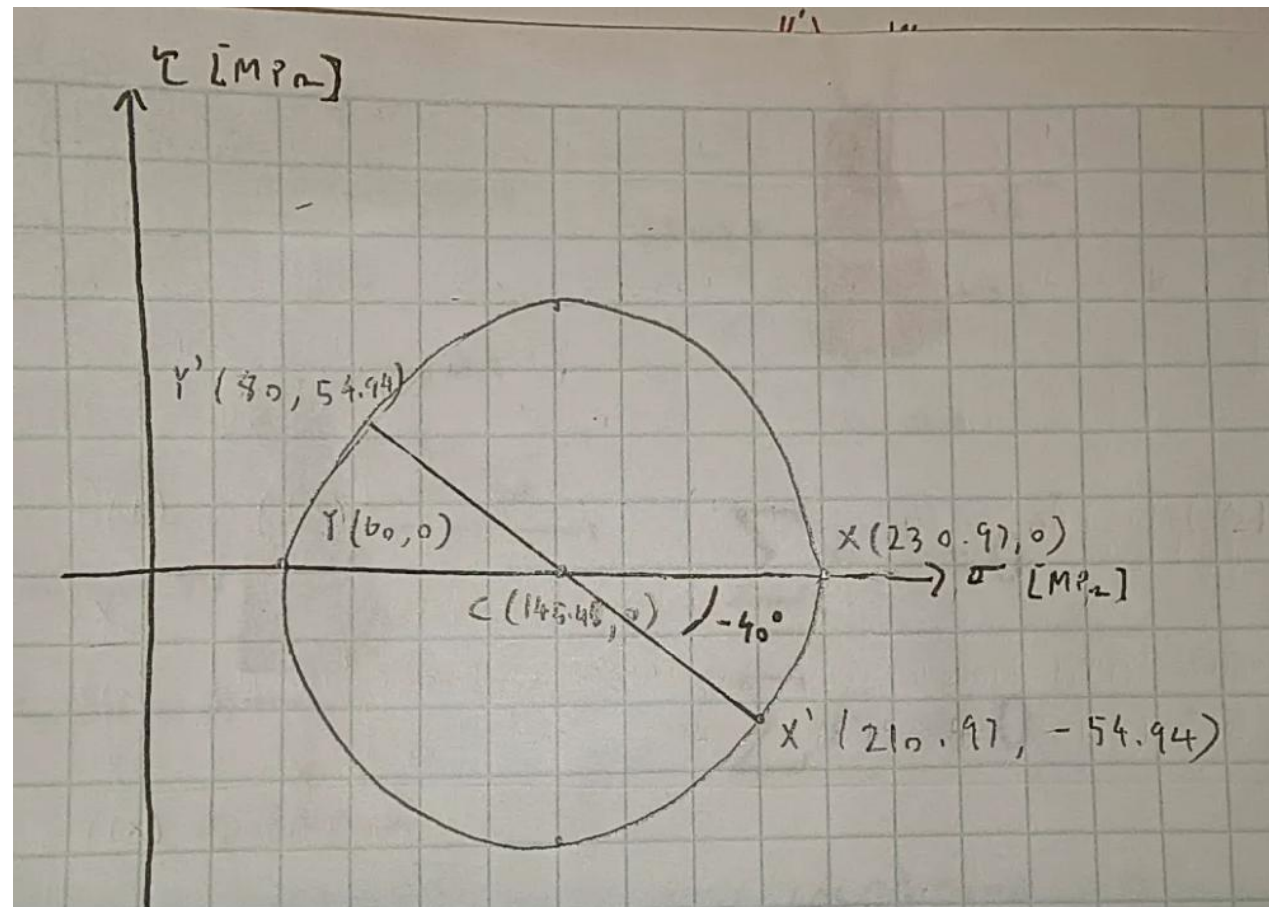
$$\sigma_{x'} = \sigma_{ave} + \frac{\Delta\sigma}{2} \cos 2\theta + \tau_{xy} \sin 2\theta = [145.48 \text{ MPa}] + \frac{[170.97 \text{ MPa}]}{2} \cos(2 \cdot -20^\circ) = 210.97 \text{ MPa}$$

$$\sigma_{y'} = 80 \text{ MPa}$$

$$\tau_{x'y'} = \tau_{xy} \cos 2\theta - \frac{\Delta\sigma}{2} \sin 2\theta = -\frac{[170.97 \text{ MPa}]}{2} \sin(2 \cdot -20^\circ) = 54.94 \text{ MPa}$$

PART B: Given $\sigma_{all,y'} = 80 \text{ MPa}$

Mohr's circle sketch



Since the first case resulted in a smaller applied stress $\sigma_x = \sigma$, it would be the limiting case. This makes sense since the cut plane is at a shallow angle such that the cut plane would see a faster increase in shear stress compared to normal stress with an increase in σ_x . [Time spent: 45 mins]

(Based on Ugural and Fenster 5th edition, p1.35) A thin-walled cylindrical pressure vessel of inner radius 120mm and wall thickness 5mm is subjected to an internal gauge pressure $p = 4$ MPa. The vessel also carries an axial compressive load $F = 30\pi$ kN and torque $T = 10\pi$ kN-m, applied through rigid end plates. Determine the maximum shearing stress and maximum and minimum normal stresses in the cylindrical wall. Show the orientation of these stresses relative to the cylinder axial and circumferential (θ) directions. Consider rotations in all planes, and sketch Mohr's circles.

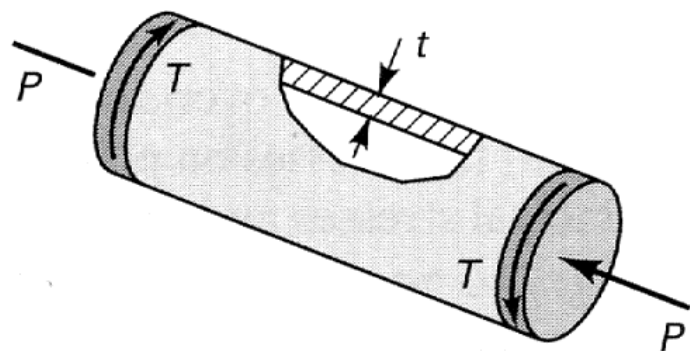
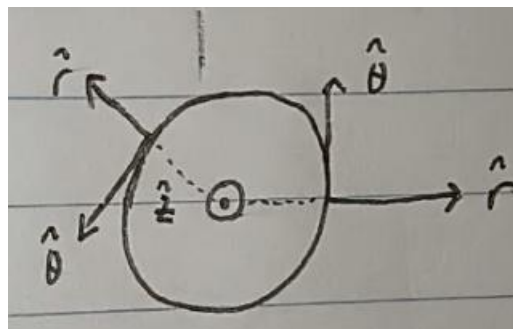
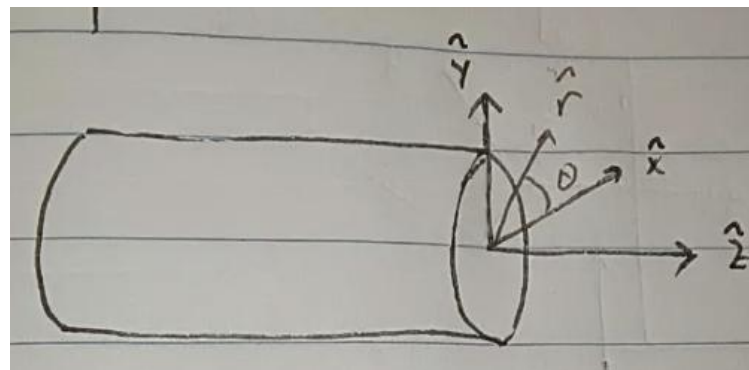


FIGURE P1.35.



Given:

- $r_i = 120$ mm
- $t = 5$ mm
- $p = 4$ MPa
- $F = 30\pi$ kN
- $T = 10\pi$ kN · m

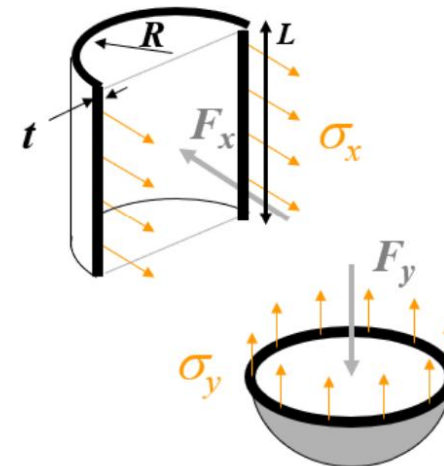
Find:

- Maximum shearing stress in the cylindrical wall
- Max and min normal stress in the cylindrical wall
- Show stresses relative to axial $\hat{z}$ and circumferential $\hat{\theta}$ directions
- Consider rotations in all planes and sketch Mohr's circles

Approach:

First determine the various stress loadings:

- Stress due to pressure (ignore the subscripts in the diagram)
 - Circumferential hoop stress $\sigma_{\theta,p} = \frac{pr_i}{t}$
 - Axial stress $\sigma_{z,p} = \frac{pr_i}{2t}$
- Stress due to applied compressive force F
 - Axial stress $\sigma_{z,F} = -\frac{F}{A_{wall}}$, $A_{wall} \approx 2\pi r_i t$
- Stress due to torsion
 - Shear stress in a thin walled hollow shaft $\tau_{z\theta} = \frac{T}{2ta}$
 - Where a is the area enclosed by the midline of the cross section
$$a = \pi \left(r_i + \frac{t}{2} \right)^2$$
- Radial stress due to pressure
 - Inner surface (compressive) $\sigma_{r,p}(r_i) = -p$
 - Outer surface (zero pressure) $\sigma_{r,p}(r_i + t) = 0$
 - Calculating for a stress element on the outer surface of the cylinder where torsional stress is greatest, we can neglect radial stress due to pressure



Approach:

Draw the 3d stress element with faces normal to the $\hat{r}, \hat{\theta}, \hat{z}$ axes. Draw the stresses computed previously. Next, identify the 3 principal axes (and principal stresses):

- By definition, a principal axis exists when there is a face normal to it that carries no shear stress (this is $\hat{r}$) since all the shear stresses exist in the $\hat{\theta} - \hat{z}$ plane, yielding a principal stress $\sigma_a = \sigma_r = 0$.
- We will then rotate the stress element in the $\hat{\theta} - \hat{z}$ plane about the $\hat{r}$ axis to find the two other principal axes. We will do this by using the formula for finding the principal angle for plane stresses (given in terms of xy as a rotation about the z axis):

$$\tan 2\theta_P = \frac{2\tau_{xy}}{\Delta\sigma}, \quad \Delta\sigma = \sigma_x - \sigma_y$$

- Finally, using θ_P , find the other two principal stresses (σ_b, σ_c) via stress transformation (given in terms of xy):

$$\sigma_{x'} = \sigma_{ave} + \frac{\Delta\sigma}{2} \cos 2\theta_P + \tau_{xy} \sin 2\theta_P$$

$$\sigma_{y'} = \sigma_{ave} - \frac{\Delta\sigma}{2} \cos 2\theta_P - \tau_{xy} \sin 2\theta_P$$

- Plot the three principal stresses and construct three Mohr's circles: $\sigma_1 \geq \sigma_2 \geq \sigma_3$
- Find max shear from max radius of Mohr's circle

Note:

This process is very similar to finding the two principal stresses for plane stress, but instead of assuming the z-axis is a principal axis, we have to first identify a shear-free coordinate axis ($\hat{r}$ in this case) to serve as our axis of rotation.

Determine Stresses

Hoop Stress:

$$\sigma_{\theta,p} = \frac{pr_i}{t} = \frac{[4 \text{ MPa}][120 \text{ mm}]}{[5 \text{ mm}]} = 96 \text{ MPa}$$

Axial Stress from Pressure:

$$\sigma_{z,p} = \frac{[4 \text{ MPa}][120 \text{ mm}]}{2[5 \text{ mm}]} = 48 \text{ MPa}$$

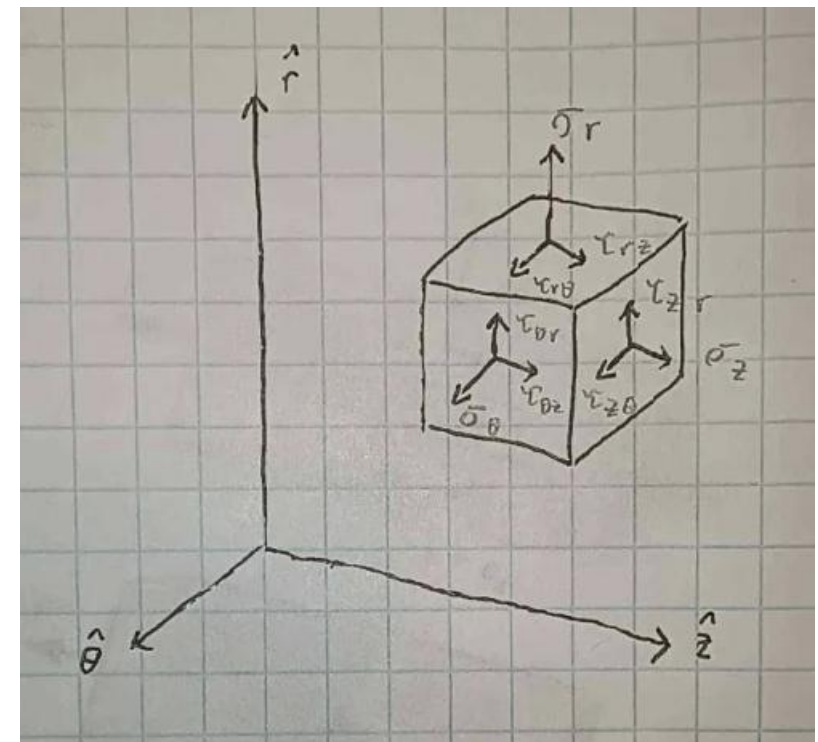
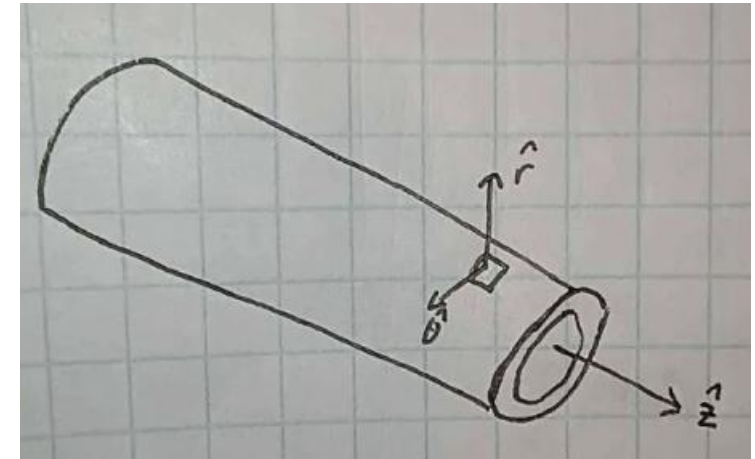
Stress due to applied compressive force F:

$$\sigma_{z,F} = -\frac{F}{2\pi r_i t} = -\frac{[30\pi \times 10^3 \text{ N}]}{2\pi[0.12 \text{ m}][0.005 \text{ m}]} = -25 \text{ MPa}$$

Shear Stress due to torsion:

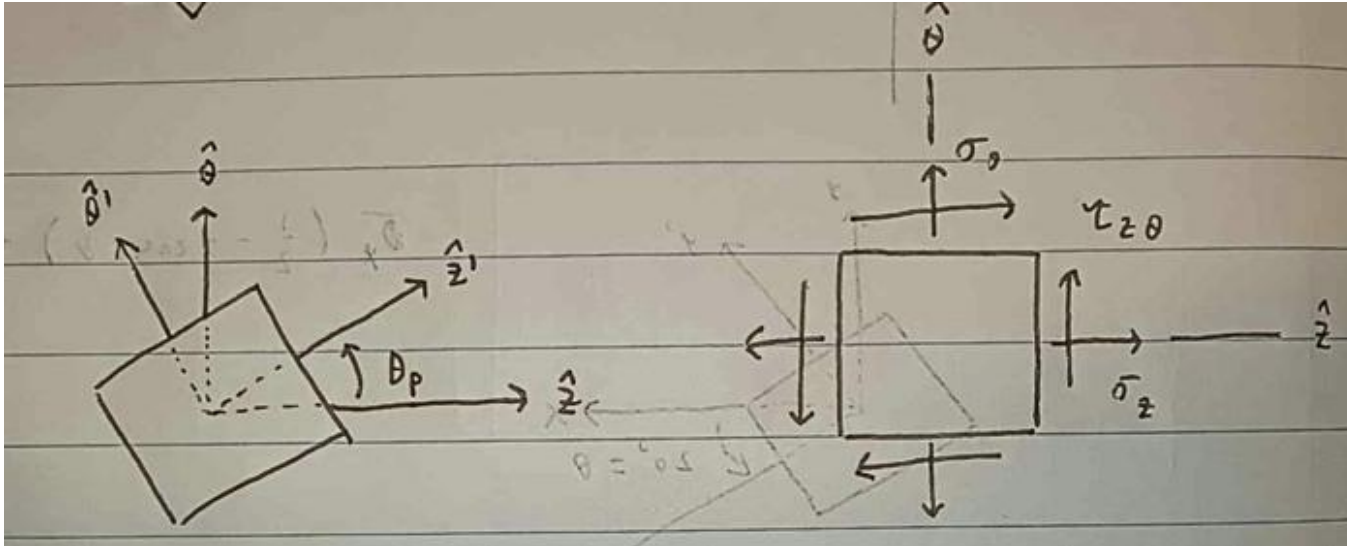
$$\tau_{z\theta} = \frac{T}{2\pi t \left(r_i + \frac{t}{2}\right)^2} = \frac{[10\pi \times 10^3 \text{ N} \cdot \text{m}]}{2\pi[0.005 \text{ m}] \left(\left[0.12 + \frac{0.005}{2}\right] \text{ m}\right)^2} = 66.63 \text{ MPa}$$

3D state of stress



Plane Stress: Since $\tau_{r\theta}, \tau_{rz}, \tau_{\theta r}, \tau_{zr} = 0$, this simplifies to a plane stress problem to find the other two principal stresses:

Principal Stress 1 (in $\hat{r}$): $\sigma_a = \sigma_r = 0$



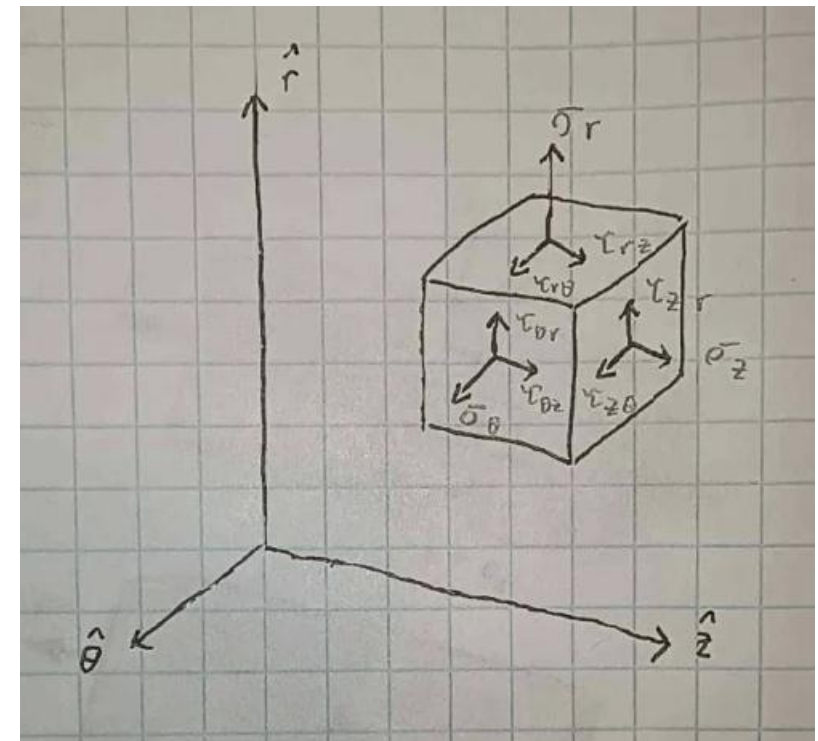
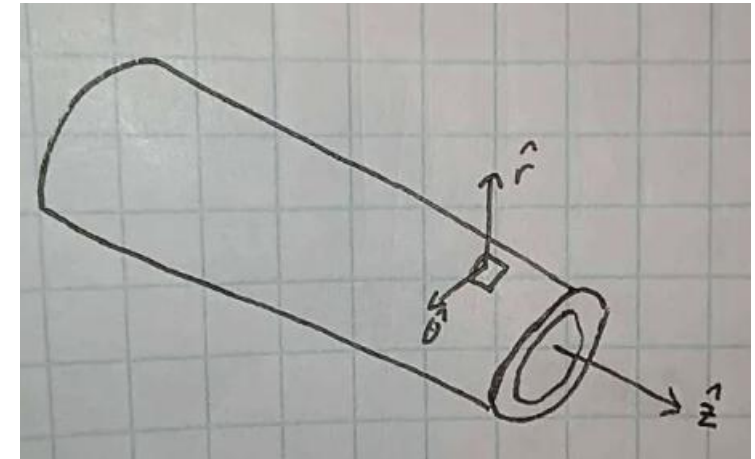
$$\sigma_z = \sigma_{z,p} + \sigma_{z,F} = [48 \text{ MPa}] + [-25 \text{ MPa}] = 23 \text{ MPa}$$

$$\sigma_\theta = \sigma_{\theta,p} = 96 \text{ MPa}$$

$$\tau_{\theta z} = \tau_{z\theta} = 66.63 \text{ MPa}$$

$$\tan 2\theta_p = \frac{2\tau_{z\theta}}{\Delta\sigma}, \quad \Delta\sigma = \sigma_z - \sigma_\theta$$

$$\tan 2\theta_p = \frac{2[66.63 \text{ MPa}]}{[23 - 96 \text{ MPa}]} \longrightarrow \theta_p = -30.64^\circ$$



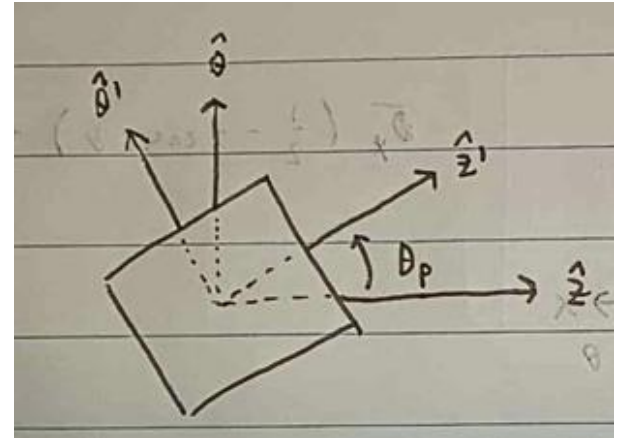
Plane Stress: Since $\tau_{r\theta}, \tau_{rz}, \tau_{\theta r}, \tau_{zr} = 0$, this simplifies to a plane stress problem to find the other two principal stresses:

$$\Delta\sigma = \sigma_z - \sigma_\theta = [23 \text{ MPa} - 96 \text{ MPa}] = -73 \text{ MPa}$$

$$\sigma_{ave} = \frac{\sigma_z + \sigma_\theta}{2} = \frac{[23 \text{ MPa} + 96 \text{ MPa}]}{2} = 59.5 \text{ MPa}$$

$$\begin{aligned}\sigma_{z'} &= \sigma_{ave} + \frac{\Delta\sigma}{2} \cos 2\theta_P + \tau_{z\theta} \sin 2\theta_P \\ &= [59.5 \text{ MPa}] + \frac{[-73 \text{ MPa}]}{2} \cos(2 \cdot -30.64^\circ) + [66.63 \text{ MPa}] \sin(2 \cdot -30.64^\circ) \\ &= -16.47 \text{ MPa} = \sigma_b\end{aligned}$$

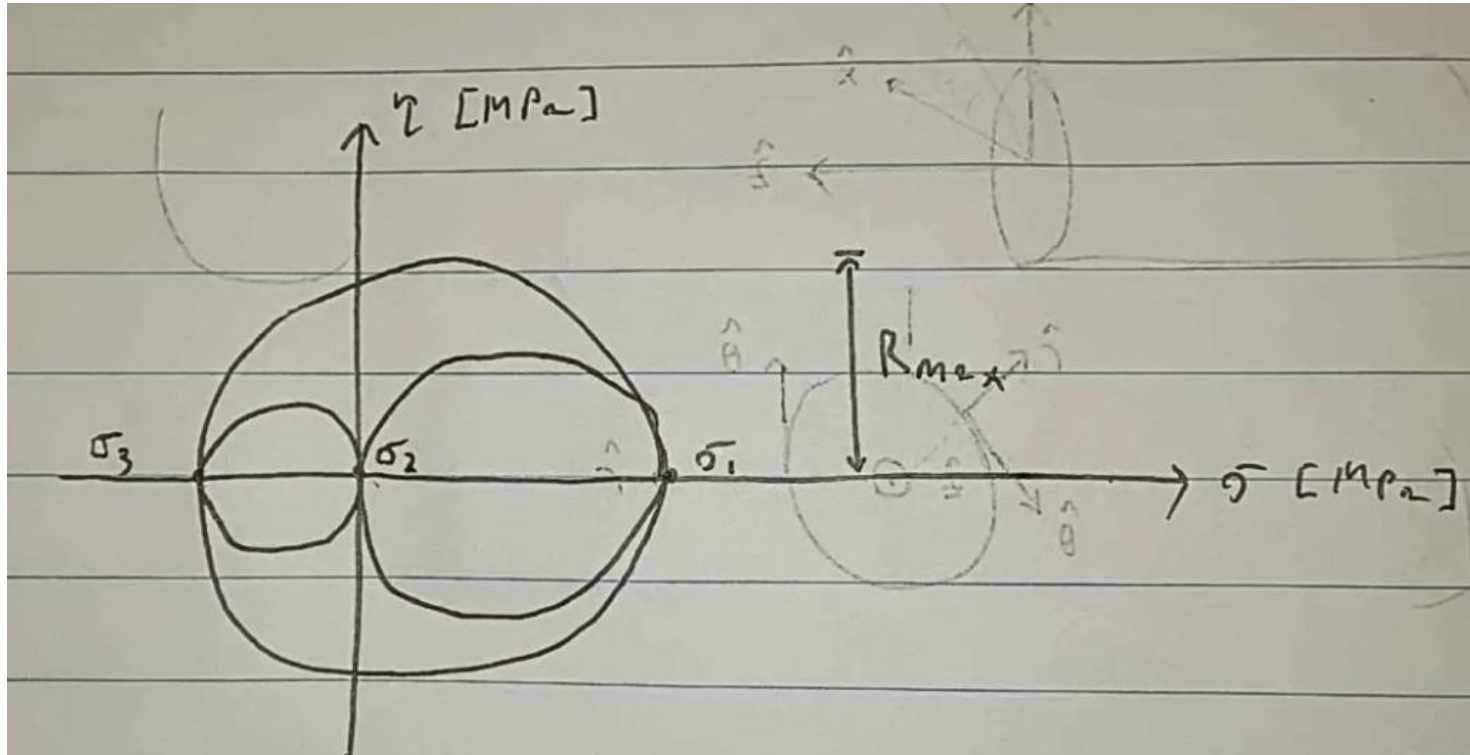
$$\begin{aligned}\sigma_{\theta'} &= \sigma_{ave} - \frac{\Delta\sigma}{2} \cos 2\theta_P - \tau_{xy} \sin 2\theta_P \\ &= [59.5 \text{ MPa}] - \frac{[-73 \text{ MPa}]}{2} \cos(2 \cdot -30.64^\circ) - [66.63 \text{ MPa}] \sin(2 \cdot -30.64^\circ) \\ &= 135.47 \text{ MPa} = \sigma_c\end{aligned}$$



3 Principal Stresses: $\sigma_a = 0$, $\sigma_b = -16.47 \text{ MPa}$, $\sigma_c = 135.47 \text{ MPa}$

Drawing 3D Mohr's Circle

$$\sigma_b < \sigma_a < \sigma_c \longrightarrow \sigma_3 = \sigma_b = -16.47 \text{ MPa}, \quad \sigma_2 = \sigma_a = 0, \quad \sigma_1 = \sigma_c = 135.47 \text{ MPa}$$



$$\text{Max normal stress: } \sigma_1 = 135.47 \text{ MPa}$$

$$\text{Min normal stress: } \sigma_3 = -16.47 \text{ MPa}$$

Max Shear Stress

$$\tau_{max} = R_{max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{[135.47 - -16.47 \text{ MPa}]}{2}$$

$$\tau_{max} = 75.97 \text{ MPa}$$

Since the max normal stress is along the $\hat{\theta}'$ direction, this means that a potential mode of failure is along an inclined plane along the surface of the cylinder, like a helix. The minimum normal stress is compressive, which makes sense since the compressive axial load and the shear from torsion combine to create net compression along the $\hat{z}'$ direction. [Time spent: 1 hr 30 mins]